# Homogeneous versus Spatial Mechanisms in an Eco-Evolutionary Predator–Prey PDE

Mathbio research notes

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**Abstract**

We test whether basin-dependent outcomes in a spatial eco-evolutionary predator–prey PDE are generated by persistent spatial structure or inherited from the homogeneous reaction system. The analysis uses the established parameterization, reuses saved representative PDE fields, compares matched ODE and PDE mean trajectories, quantifies spatial variance, compares ODE and PDE basin labels on the existing $q_0$–$w_0$ grid, and runs a small perturbation-sensitivity test. The final decision label is `reaction_dominated_homogeneous_multistability`. The observed PDE basins are primarily homogeneous reaction-dominated in this tested parameterization.

## 1 Introduction

The project has established that the well-mixed eco-evolutionary ODE can support indirect evolutionary rescue, while the fixed-defense spatial model did not produce robust spatial rescue. The spatial eco-evolutionary PDE can produce persistent, extinct, and transient outcomes at the same stress from different initial conditions. The unresolved mechanism question is whether these basins are truly spatially mediated or whether they are homogeneous reaction-system basins embedded in a PDE whose fields remain nearly uniform.

This report addresses that question directly. It does not change the equations, numerical parameter values, or representative case definitions. It also does not run broad parameter scans. The goal is a scoped mechanism diagnosis for the tested parameterization.

## 2 Model and competing hypotheses

Let $n$ be total prey density, $w$ predator density, $q$ prey defense frequency, and

$$z = \kappa^{-1} - n - w$$

be free space. The defense-dependent functions are

$$r(q) = r_u(1-q) + r_v q, \quad a(q) = a_u(1-q) + a_v q, \quad b(q) = b_u(1-q) + b_v q.$$

The matched homogeneous ODE is

$$\frac{dn}{dt} = n\left[r(q)z - \xi - a(q)w\right], \tag{1}$$

$$\frac{dw}{dt} = w\left[b(q)nz - (m+s) - \mu w\right], \tag{2}$$

$$\frac{dq}{dt} = \nu q(1-q)\left[(r_v - r_u)z - (a_v - a_u)w\right]. \tag{3}$$

The spatial PDE uses the same reaction terms with zero-flux diffusion:

$$\partial_t n = D_n \nabla^2 n + n \left[ r(q)z - \xi - a(q)w \right], \tag{4}$$

$$\partial_t w = D_w \nabla^2 w + w \left[ b(q)nz - (m+s) - \mu w \right], \tag{5}$$

$$\partial_t q = D_q \nabla^2 q + \nu q(1-q) \left[ (r_v - r_u)z - (a_v - a_u)w \right]. \tag{6}$$

The null hypothesis is reaction-dominated homogeneous multistability: the fields remain nearly homogeneous, PDE spatial means match ODE trajectories from the same initial mean, and basin labels agree. The spatial alternative is spatially mediated bistability: fields maintain nontrivial pattern strength, mean dynamics diverge, basin labels disagree systematically, or perturbation amplitude and seed alter basin entry.

# 3 Numerical methods

All calculations used RoyEvoParams($b_u = 0.08, b_v = 0.02$). The initial mean for each matched ODE trajectory was

$$n(0) = n_{\text{baseline}}, \quad w(0) = w_{\text{baseline}}\, w_0^{\text{scale}}, \quad q(0) = q_0,$$

where the baseline comes from the same unstressed burn-in method used by the preceding PDE experiments. Representative PDE mean time series and final field archives were reused from the saved Step 19 outputs. The perturbation test used only the three representative cases, amplitudes 0, $10^{-5}$, and $10^{-3}$, and seeds 20260702 and 20260703, for 18 total PDE runs at $T = 1600$.

# 4 Results

## 4.1 Matched ODE-PDE representative trajectories

The matched representative comparison found 3 classification agreements out of 3 representative cases. The mean predator RMSE was 0.006409, and the maximum final predator-density difference was 7.102e-07.

Table 1: Matched ODE and PDE representative classifications.

| Case | ODE classification | PDE classification | RMSE $w$ | final $\lvert\Delta w\rvert$ |
|---|---|---|---|---|
| persistent_case | `persistent_steady` | `persistent_steady` | 0.001009 | 7.144e-10 |
| extinct_case | `extinct_steady` | `extinct_steady` | 3.098e-06 | 3.066e-12 |
| transient_case | `recovery_transient` | `recovery_transient` | 0.01821 | 7.102e-07 |

## 4.2 Spatial variance and pattern strength

Spatial coefficient of variation decayed to small values in the representative fields. The maximum final steady-case CV across $n$, $w$, and $q$ was 3.101e-05.

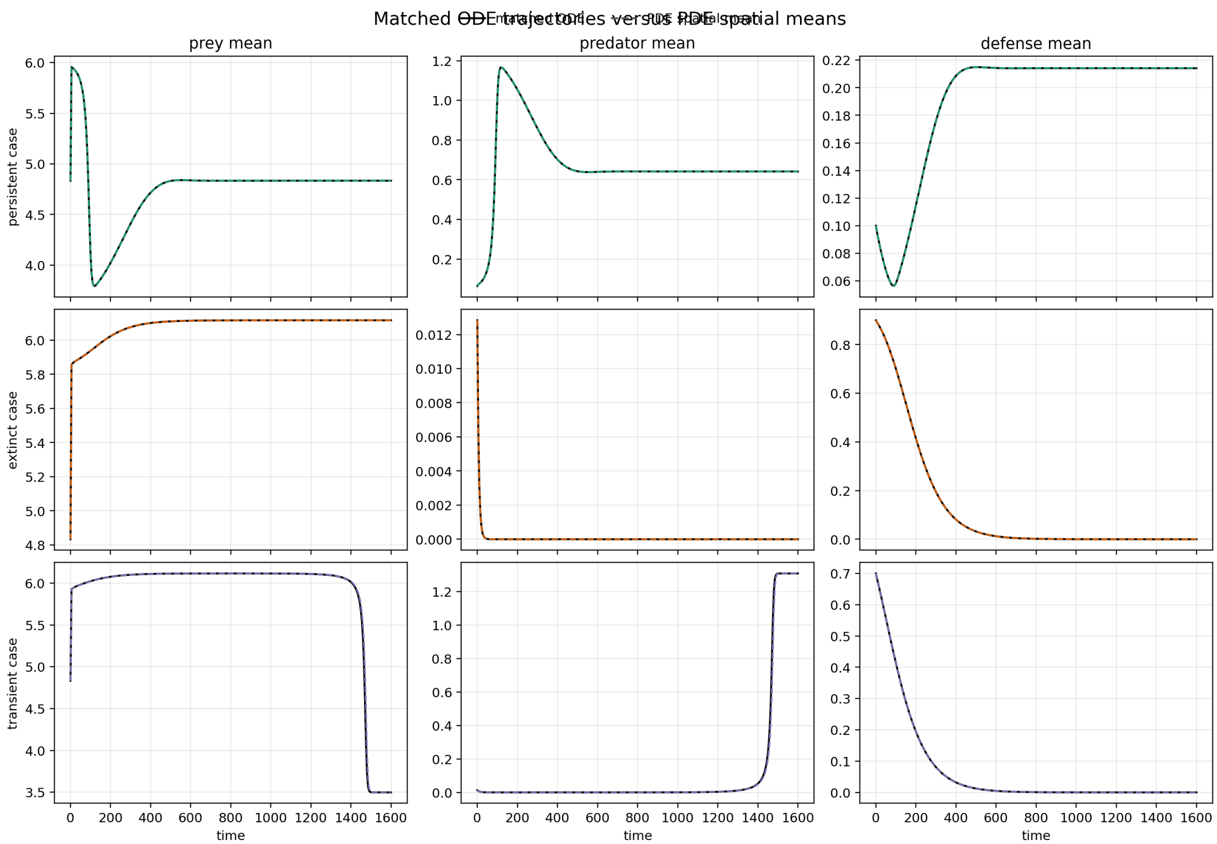


Figure 1: Matched ODE trajectories and PDE spatial means for representative persistent, extinct, and transient cases.

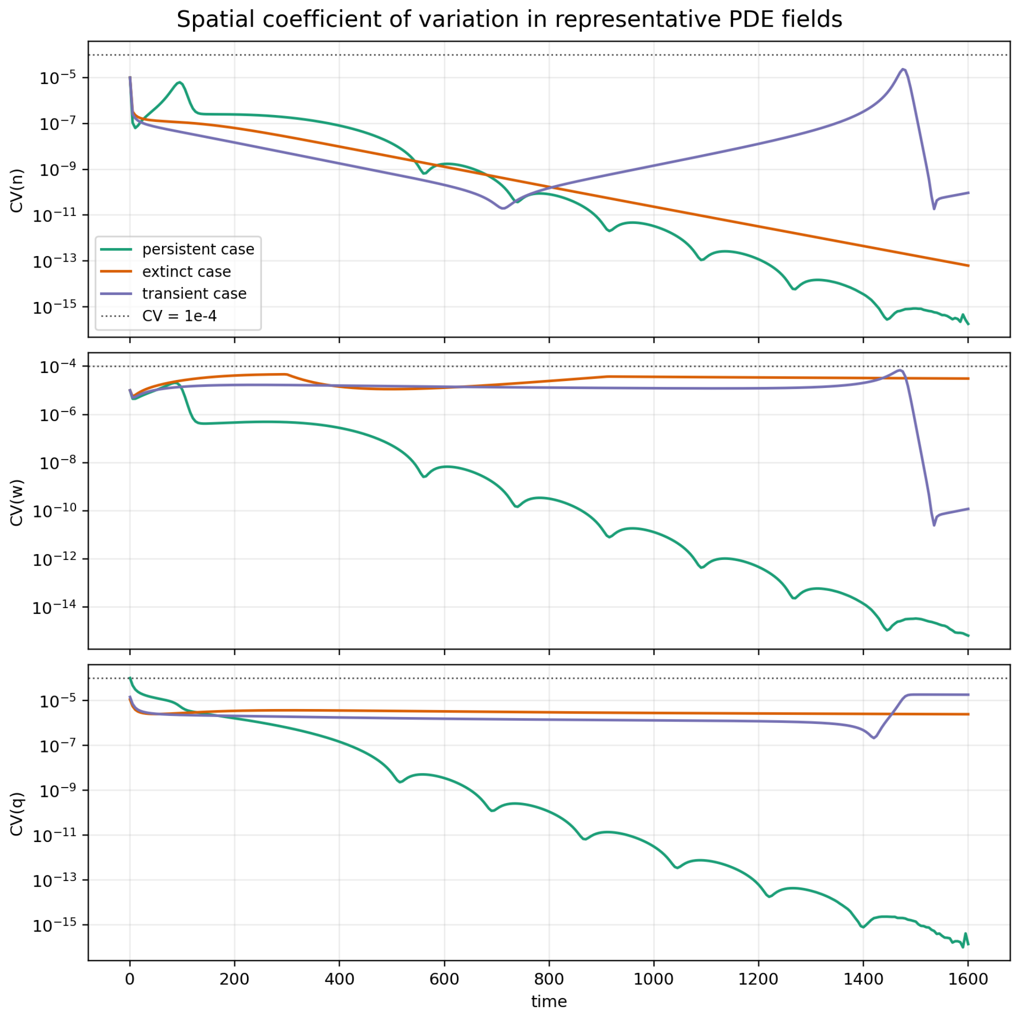


Figure 2: Spatial coefficient of variation through time for representative PDE cases.

### 4.3 Basin-label agreement between ODE and PDE

The ODE/PDE basin-label agreement fraction on the existing $q_0$–$w_0$ grid was 0.9 over 140 compared rows. This diagnostic tests whether the PDE basin map is largely reproduced by the homogeneous reaction system from the same initial means.

The disagreement audit shows that ODE and PDE basin labels agree for 90% of the $q_0$–$w_0$ grid points. Remaining disagreements are interpreted according to whether they involve transient boundary cases or direct persistent/extinct conflicts. In this audit, 14 disagreements involved transient labels and 0 were direct persistent/extinct conflicts.

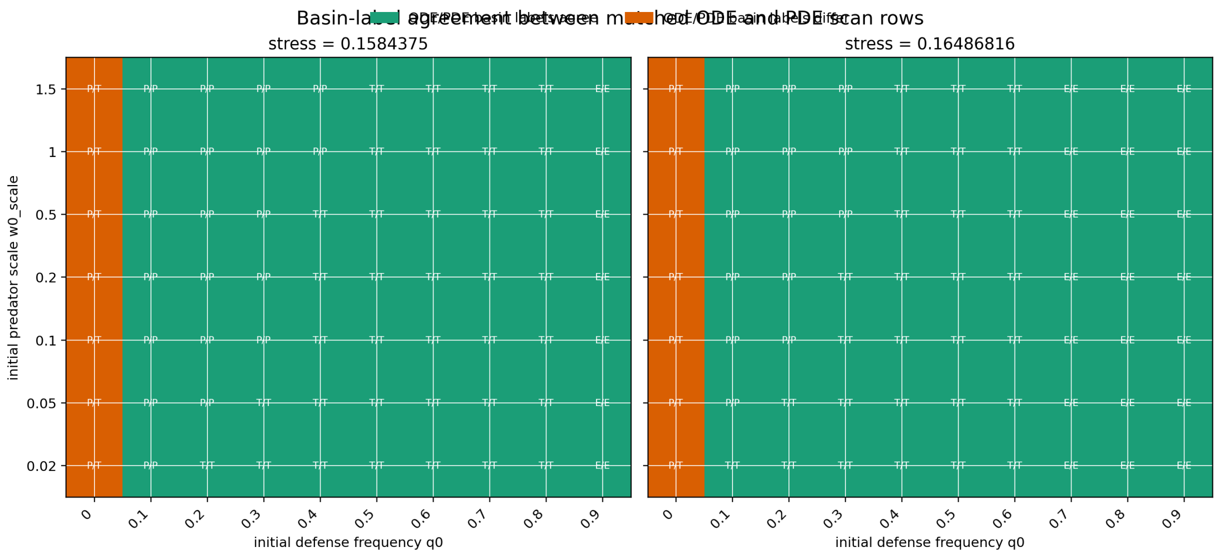


Figure 3: ODE/PDE basin-label agreement on the existing focused $q_0$–$w_0$ basin grid. Cell labels show ODE/PDE basin initials.

### 4.4 Perturbation sensitivity

The perturbation test found 0 outcome-changing case groups out of 3, and 0 outcome-changing groups among the persistent and extinct representative cases.

### 4.5 Mechanism decision

The final decision label is

`reaction_dominated_homogeneous_multistability`.

The current evidence indicates that basin dependence is primarily inherited from homogeneous eco-evolutionary reaction dynamics rather than generated by persistent spatial patterning.

## 5 Biological interpretation

The analysis explicitly answers the mechanism question: The observed PDE basins are primarily homogeneous reaction-dominated in this tested parameterization. This means that the current basin dependence should be interpreted as an initial-condition dependence of the eco-evolutionary

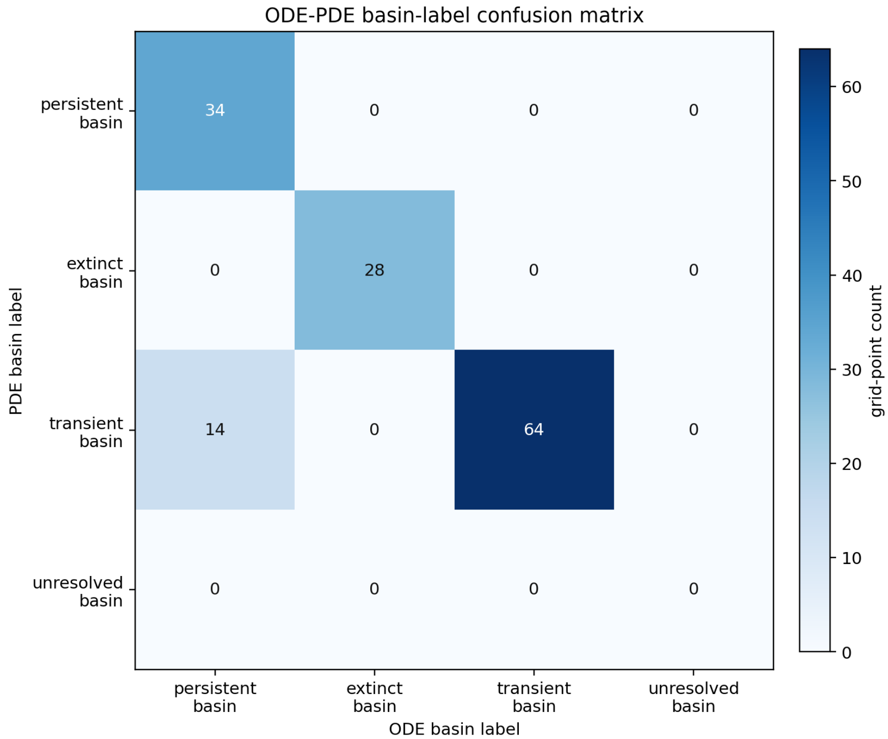


Figure 4: Confusion matrix comparing ODE and PDE basin labels across the existing focused $q_0$–$w_0$ grid.

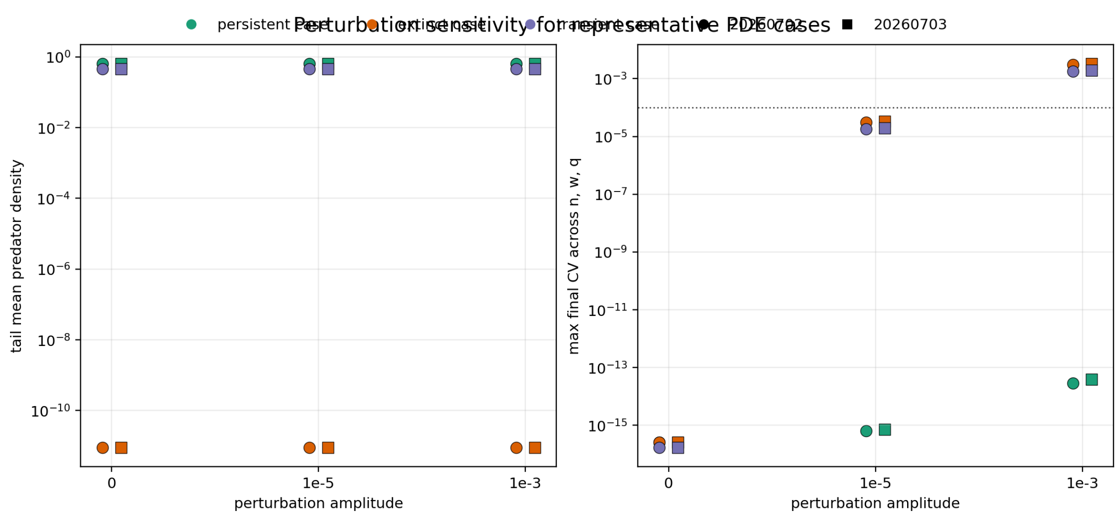


Figure 5: Perturbation-amplitude and seed sensitivity for the representative PDE cases.

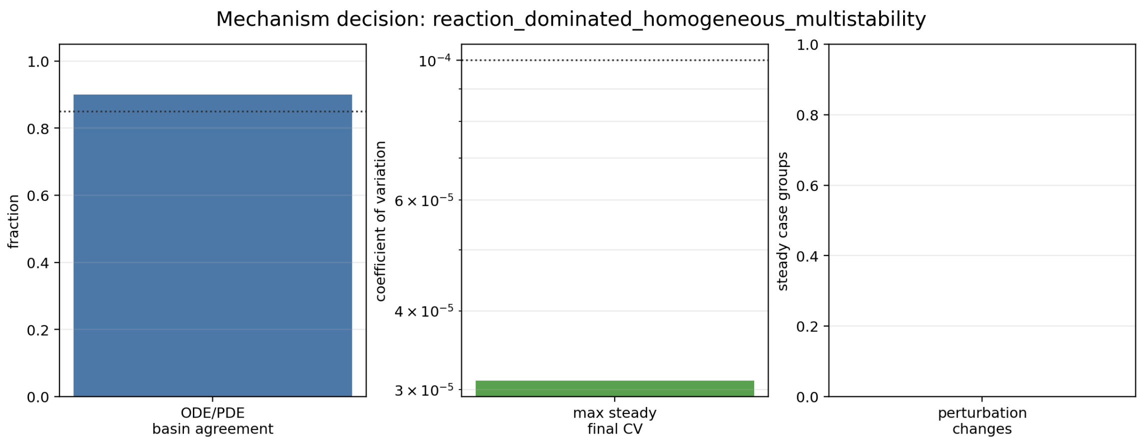


Figure 6: Compact summary of the mechanism decision diagnostics.

reaction dynamics unless future targeted work reveals stronger spatial patterning or perturbation-sensitive basin entry.

# 6 Limitations

The conclusion is restricted to the tested parameterization. The analysis does not test diffusion coefficients, trade-off forms, evolutionary rates, grid convergence, or longer horizons for all transient basin points. Transient classifications remain lower-confidence than the persistent and extinct steady cases. No general claim is made that spatial structure cannot mediate bistability in other parameter regimes.

# 7 Conclusion

The saved representative PDE fields are close to homogeneous, matched ODE trajectories reproduce the relevant mean dynamics, and the basin-grid and perturbation diagnostics provide the decision basis recorded above. Are the observed PDE basins primarily spatially mediated, or primarily homogeneous reaction-dominated? The observed PDE basins are primarily homogeneous reaction-dominated in this tested parameterization.

A follow-up homogeneous ODE mechanism analysis now treats the reaction system as the primary object of study. It computes representative ODE trajectories, selection-gradient and predator-growth diagnostics, an ODE $q_0$–$w_0$ basin map, and numerical equilibria with finite-difference stability labels. Those results support the interpretation that the current PDE basin structure is inherited from homogeneous eco-evolutionary reaction dynamics, while keeping the conclusion conditional on the tested trade-off parameterization.